

Satellite Image Land-Use Classifier & Temporal Change Detector

Project Report

1. Overview

This project builds a computer vision system that (1) classifies land-use types from satellite imagery using transfer learning, and (2) detects land-cover change between two time periods using embedding-based similarity. The system is delivered as a set of reproducible notebooks and a working interactive Streamlit dashboard.

2. Datasets

Dataset	Size	Classes	Role	Source
EuroSAT	27,000 tiles	10	Train / Validation	Sentinel-2, 10m/pixel
UC Merced Land Use	2,100 images	21 (8 mapped)	Holdout / cross-domain test	USGS aerial, ~0.3m/pixel

3. Problem Statement

Manual monitoring of land-use change from satellite imagery is slow and expensive. This project explores whether a lightweight, transfer-learning-based pipeline can (a) reliably classify land-use type from a single tile and (b) flag likely change between two tiles of the same location, using only embedding similarity - without any explicit change-detection labels.

4. Base Model & Deliverables

Base model: ResNet-18 (ImageNet pretrained), fine-tuned with a two-phase strategy. **Deliverables:** data pipeline with spatial block split, scratch baseline CNN, fine-tuned classifier, embedding-based change detector with ROC-justified threshold, Streamlit dashboard, spatial leakage experiment, error analysis, and this report. All four bonus tasks (GradCAM, multi-threshold toggle, embedding t-SNE comparison, class imbalance experiment) are also implemented and reported below.

5. Methodology

5.1 Data Pipeline & Spatial Block Split

EuroSAT tiles are cropped from a small number of larger source scenes, so images from the same area can appear as near-duplicates. A naive random train/val/test split can therefore place near-duplicate tiles on both sides of a split, inflating validation performance through data leakage. To avoid this, images were grouped into pseudo-spatial *blocks* (same-class images grouped by filename order, ~9 images per block) and whole blocks - not individual images - were assigned to train/val/test at a 70/15/15 ratio. A second, naive random split was also built and used *only* for the leakage comparison in Section 8 - never for training any reported model.

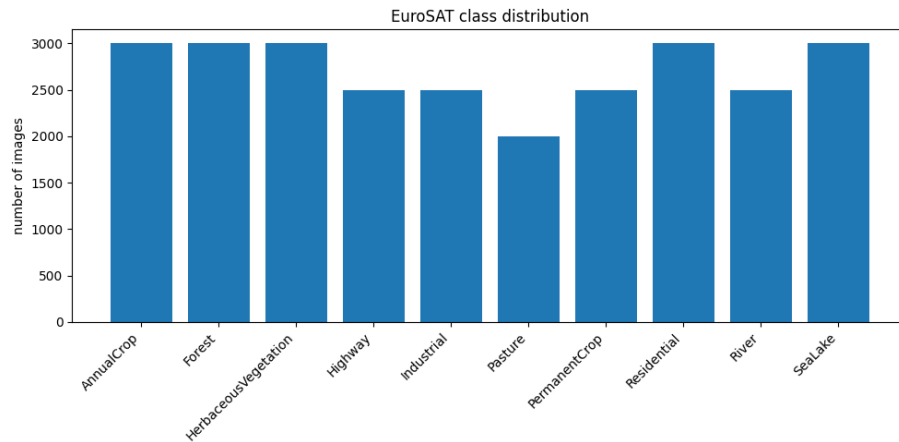


Figure 1. EuroSAT class distribution (train split). Classes are reasonably balanced, which is the basis for the deliberate imbalance experiment in Section 9.

5.2 Module 1 - Classifier Architecture

A 3-layer scratch CNN (conv-ReLU-pool x3, dropout 0.3) was trained first as a baseline floor. The main classifier is a ResNet-18 (ImageNet pretrained) fine-tuned in two phases: **Phase 1** - backbone frozen, only the replaced 10-class head trained, 3 epochs, LR 1e-3. **Phase 2** - the last two convolutional blocks (layer3, layer4) unfrozen, LR reduced 10x to 1e-4, trained 5 more epochs. ImageNet normalization was applied for all ResNet-18 training, matching the pretrained weights' expected input distribution.

5.3 Module 2 - Change Detection Approach

The fine-tuned ResNet-18's classification head was replaced with an identity layer, turning it into a 512-dim feature extractor. EuroSAT has no real timestamps, so a before/after (T1/T2) time series was *simulated*: each spatial block was split in half, and a random 50% of T2 images were deliberately swapped for an image from a different class to create labeled ground-truth "changed" pairs; the remainder are genuine same-class "unchanged" pairs. Cosine similarity between T1/T2 embeddings is the change signal; an operating threshold was selected using Youden's J statistic (max of TPR-FPR) on the resulting ROC curve.

6. Module 1 Results - Land-Use Classifier

6.1 Baseline vs. Fine-tuned Comparison

Model	EuroSAT Val Accuracy	EuroSAT Val Macro-F1
Baseline CNN (scratch, 3-layer)	78.3%	0.779
Fine-tuned ResNet-18 (two-phase)	95.7%	0.956

6.2 Frozen vs. Unfrozen (Phase 1 vs Phase 2)

Phase	Trainable Params	Final Val Accuracy
Phase 1 (frozen backbone)	5,130 / 11.18M	85.2%
Phase 2 (last 2 blocks unfrozen)	10,498,570 / 11.18M	95.7%

Unfreezing the last two convolutional blocks and dropping the learning rate 10x nearly doubled trainable parameters and lifted val accuracy by ~10.5 points over the frozen-only baseline.

6.3 Training Curves & Confusion Matrix

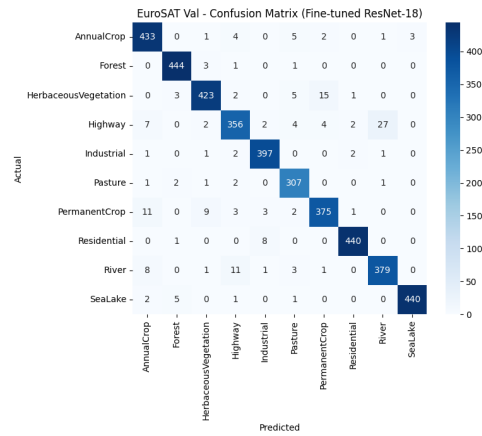
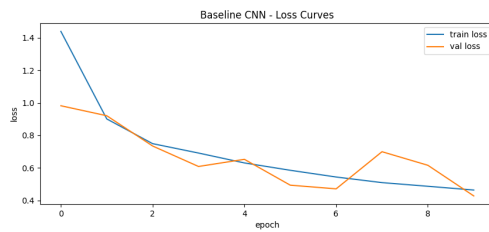


Figure 2 (left). Baseline CNN train/val loss over 10 epochs - mild overfitting visible after epoch 6. Figure 3 (right). Fine-tuned ResNet-18 confusion matrix on EuroSAT validation. Highway and PermanentCrop remain the most confused classes.

6.4 UC Merced Cross-Domain Holdout

UC Merced has 21 classes on a different sensor/resolution (USGS aerial, ~0.3m/pixel vs. EuroSAT's Sentinel-2, 10m/pixel). Only 8 of EuroSAT's 10 classes have a defensible conceptual match in UC Merced (PermanentCrop and SeaLake have none and were excluded, rather than force a mapping). Predictions were still scored against the full 10-class space so any spurious PermanentCrop/SeaLake predictions would be visible.

Evaluation	Accuracy
EuroSAT validation (in-domain)	95.7%
UC Merced holdout (cross-domain, 8/10 classes)	29.7%

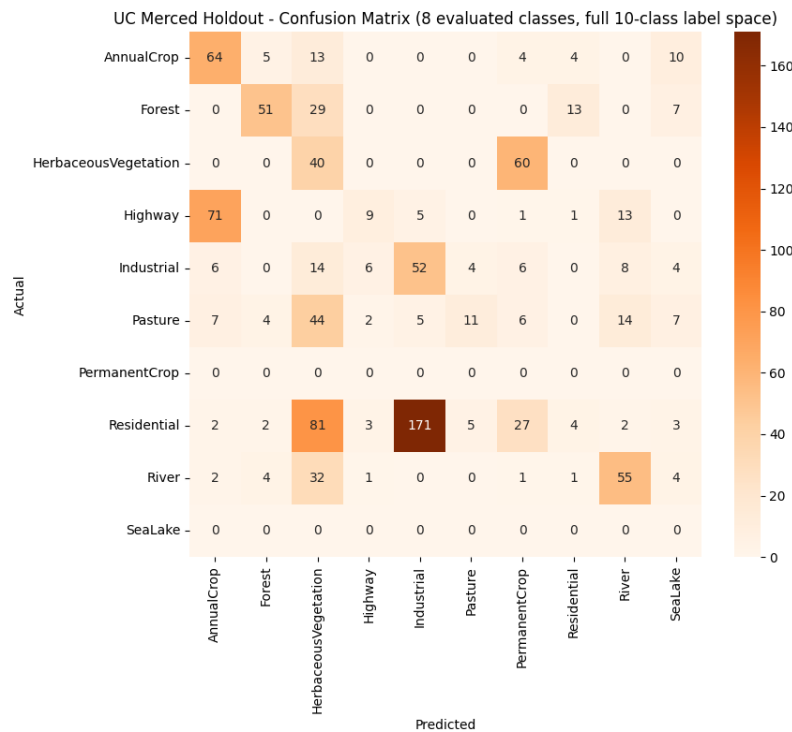


Figure 3b. Confusion matrix on the UC Merced holdout (ground truth restricted to the 8 mapped classes; prediction space is the full 10 EuroSAT classes, so spurious PermanentCrop / SeaLake predictions would be visible - none occurred).

The 66-point drop is a genuine, expected finding rather than a bug: it reflects the sensor/resolution domain gap between Sentinel-2 and aerial imagery, not a failure of the fine-tuning procedure. Residential was the single worst-performing class (only 4 of 300 correctly identified) - the confusion matrix shows it was overwhelmingly misclassified as Industrial (171 of 300)

and, secondarily, HerbaceousVegetation (81 of 300). This is a more specific and useful finding than a generic "looks different": the rooftop/built-structure texture of UC Merced's aerial residential tiles apparently resembles the model's learned Industrial pattern more than its learned (Sentinel-2-derived) Residential pattern. Highway shows a similar, on-theme confusion to Section 9's EuroSAT error analysis - 71 of 100 Highway tiles were misclassified as AnnualCrop, consistent with the same road-bordered-by-farmland confusion pattern observed in-domain.

7. Module 2 Results - Change Detection

7.1 ROC Curve & Threshold Selection

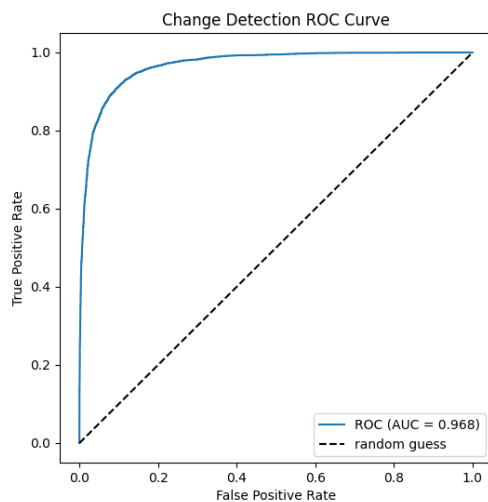


Figure 4. ROC curve for the embedding-similarity change detector (AUC ~0.97, run-to-run variation of +/-0.01 observed due to random T1/T2 pair construction).

Operating Point	Cosine Threshold	TPR	FPR
High Recall	0.486	0.970	0.180
Balanced (Youden's J)	0.443	0.931	0.094
High Precision	0.369	0.752	0.020

The balanced threshold was selected via Youden's J statistic (max of TPR-FPR), giving a reasonable trade-off (93% of real changes caught, 9% false alarm rate) rather than an arbitrary cutoff. All three operating points are exposed as a toggle in the dashboard (Bonus B).

7.2 Sample Change Heatmaps

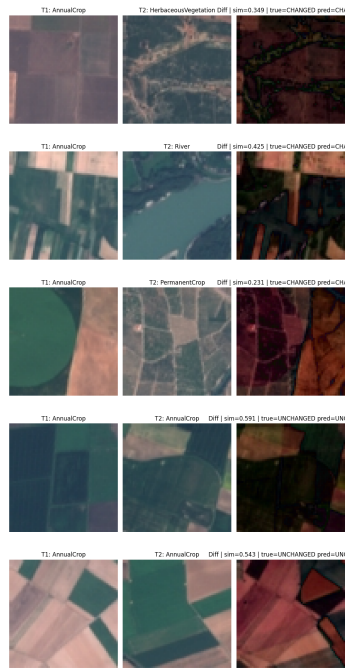


Figure 5. Five sample T1/T2 pairs with pixel-difference heatmaps, similarity score, ground truth, and predicted label at the balanced threshold. All 5 predictions match ground truth. Note: the pixel-difference panel is a visual aid only - the actual detection signal is embedding cosine similarity, not raw pixel differencing.

8. Spatial Leakage Experiment

To quantify leakage risk, the identical baseline CNN (same architecture, hyperparameters, epoch count) was trained on two different splits of the same 27,000 images: the spatial block split described in Section 5.1, and a naive random split where individual images are shuffled independently.

Split Strategy	Val Accuracy
Random split (naive)	85.7%
Spatial block split (correct)	78.3%

A 7.4-point gap from split strategy alone, with identical training data volume, model, and hyperparameters. This is consistent with data leakage: near-duplicate or spatially adjacent tiles land on both sides of the random split, letting the model partially memorize location-specific detail rather than learn generalizable patterns. The block split (used for every other result in this report) is the more honest estimate of real-world generalization, despite its lower headline number. *Caveat*: blocks are a filename-order proxy, not verified GPS coordinates, so this should be read as strong circumstantial evidence of spatial leakage rather than a rigorously proven geographic claim.

9. Error Analysis

The five highest-confidence misclassifications (all >99.99% confidence, all wrong) were pulled from the fine-tuned model's validation predictions, to surface systematic rather than borderline errors.

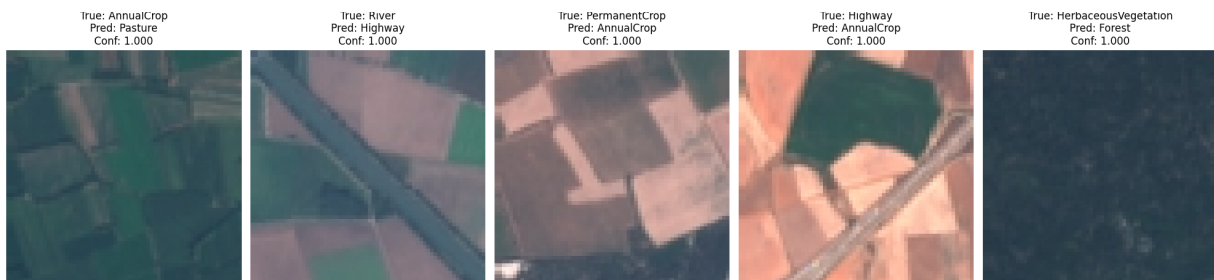


Figure 6. Top-5 highest-confidence misclassifications.

True	Predicted	Hypothesis
AnnualCrop	Pasture	Early-growth/fallow crop visually resembles open grassland at 10m/px.
River	Highway	Calm, low-contrast water can resemble a paved linear surface.
PermanentCrop	Weakest AnnualCrop	AnnualCrop overall (lowest two F1 scores); orchard/vineyard rows can resemble annual crop rows depending on growth
Highway	AnnualCrop Narrow road bordered by dominant farmland; crop pixels may out-weigh the thin road strip.	
HerbaceousVegetation	Forest	Dense herbaceous cover can blend with sparse/young forest texture.

Common thread: every error occurs between conceptually adjacent classes (crop-type or linear-feature confusions) - the model fails at fine-grained distinctions between visually similar land-use types, not at random.

10. Bonus Tasks

10.1 Bonus A - GradCAM Visualisation

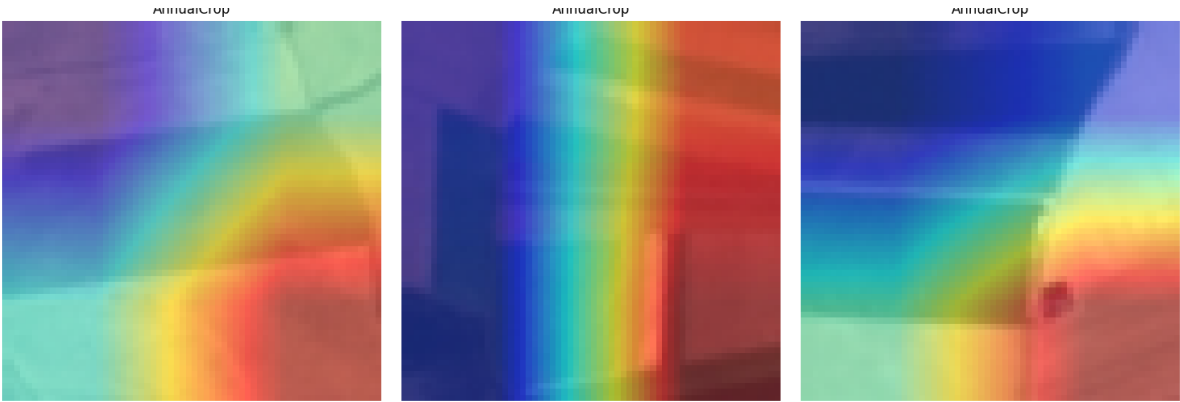


Figure 7. GradCAM overlays on the fine-tuned model's final conv block (layer4) for three sample tiles, showing which pixels most influenced the prediction.

10.2 Bonus C - Embedding Visualisation (t-SNE)

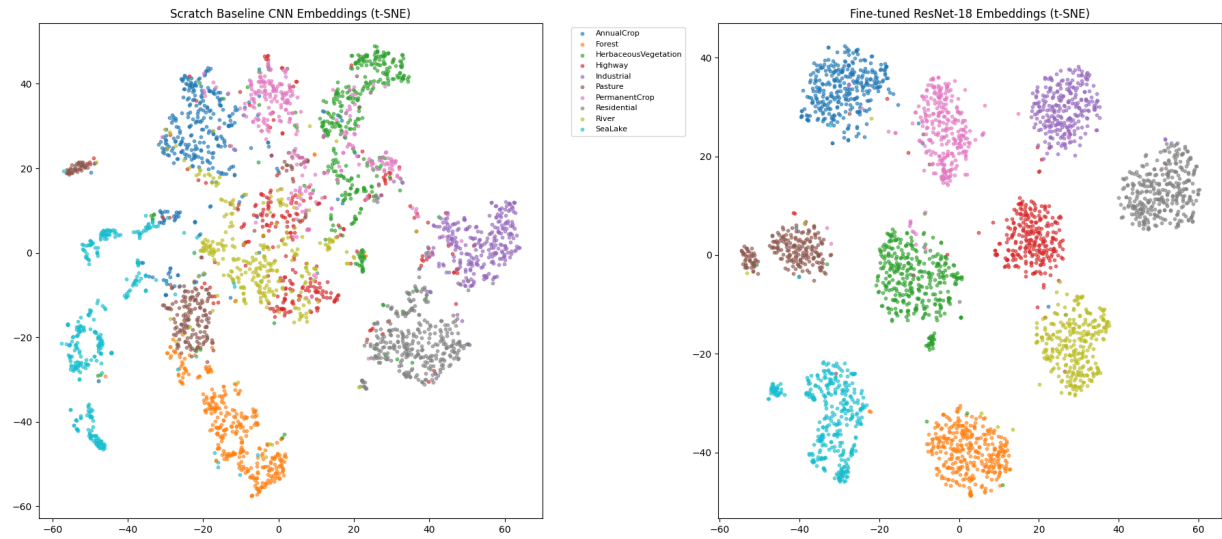


Figure 8. t-SNE projection of 3,000 sampled embeddings: scratch baseline CNN (left, 256-dim) vs. fine-tuned ResNet-18 (right, 512-dim). The fine-tuned model's classes form visibly tighter, more separated clusters, consistent with its much higher accuracy.

10.3 Bonus D - Class Imbalance Experiment

Highway and PermanentCrop (already the two weakest classes in the balanced baseline) were downsampled to 20% of their original training size (~1,700 to ~350 each), all other classes left untouched, and the baseline CNN retrained. Class-weighted cross-entropy loss (weights inversely proportional to class frequency) was then applied as the mitigation, with no other changes.

Scenario	Highway F1	PermanentCrop F1	Macro-F1 (all classes)
Balanced baseline	0.631	0.600	0.779
Imbalanced, no mitigation	0.182	0.090	0.687
Imbalanced + weighted loss	0.611	0.680	0.810

Without mitigation, both starved classes collapsed (recall as low as 0.048) and the model redirected their tiles into visually similar neighbours - consistent with Section 9's error patterns. Weighted loss alone recovered both classes close to (Highway) or above (PermanentCrop) their original balanced performance, and raised macro-F1 above even the original balanced baseline (0.810 vs 0.779) - showing weighted loss is a simple, architecture-agnostic, and effective mitigation. Notably, plain accuracy barely moved (78.3% to 74.5%) while macro-F1 dropped much more sharply (0.779 to 0.687), underscoring that accuracy alone can mask imbalance-driven failures.

10.4 Bonus B - Multi-Threshold Toggle

The Streamlit dashboard (Section 11) exposes all three ROC-derived operating points from Table in Section 7.1 (High Recall / Balanced / High Precision) as a live radio-button toggle, letting the user see the change flag shift in real time as sensitivity changes.

11. Geo-Dashboard

A Streamlit app loads the fine-tuned checkpoint once (cached), accepts two uploaded tiles, and displays predicted class + confidence for each tile, their embedding cosine similarity, a pixel-difference heatmap, and a change flag driven by the selected threshold mode. Runs entirely locally, no internet dependency after initial package install.

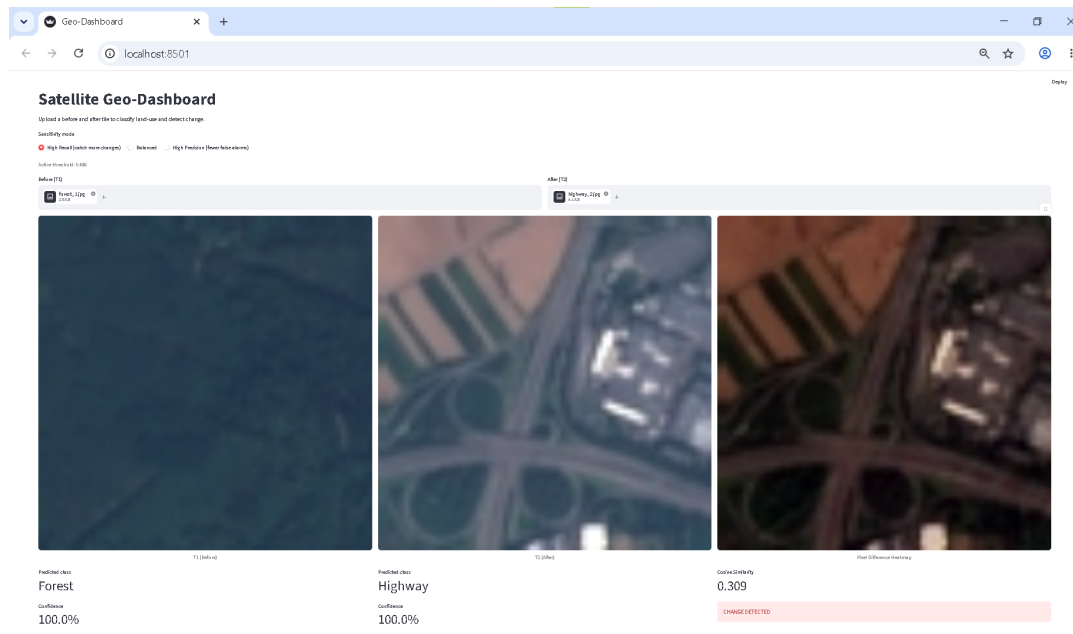


Figure 9. Dashboard, High Recall mode: Forest vs. Highway tiles correctly flagged as CHANGE DETECTED (similarity 0.309).

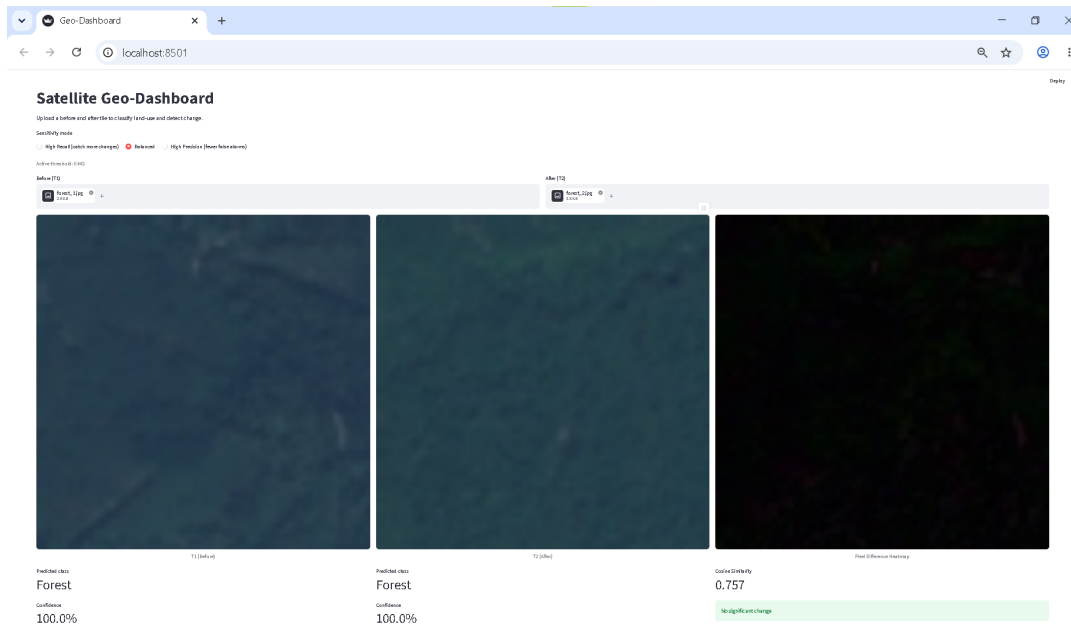


Figure 10. Dashboard, Balanced mode: two different Forest tiles correctly flagged as No significant change (similarity 0.757).

12. Limitations

- **Simulated temporal data.** EuroSAT has no real timestamps; T1/T2 pairs and "change" labels are constructed by splitting spatial blocks and artificially swapping classes, not observed over real time.
- **Spatial blocks are a filename-order proxy**, not verified GPS-based geographic grouping - the leakage finding is strong circumstantial evidence, not a rigorously proven spatial claim.
- **UC Merced coverage is partial** (8 of 10 EuroSAT classes have a defensible mapping); PermanentCrop and SeaLake were never evaluated cross-domain.
- **Large in-domain/cross-domain gap** (95.7% to 29.7%) shows the model does not generalize across sensor types without further domain adaptation or fine-tuning on target-domain data.
- **Dashboard is a local prototype** (no deployment, no batch processing, single tile-pair at a time).

13. Conclusion

Two-phase transfer learning lifted land-use classification from a 78.3% scratch-CNN floor to 95.7% (macro-F1 0.956), demonstrating the value of ImageNet pretraining even for a domain (satellite imagery) quite different from ImageNet's natural photos. The embedding-based change detector achieved a strong ROC AUC (~ 0.97) using only cosine similarity, with no dedicated change-detection training. Two supporting experiments - spatial leakage (7.4-point accuracy gap from split strategy alone) and class imbalance (macro-F1 recovered from 0.687 to 0.810 via weighted loss) - reinforce that methodology choices around data splitting and class balance can matter as much as model architecture. The largest open finding is the domain gap exposed by the UC Merced holdout, which would be the natural next direction: domain adaptation or fine-tuning on a small sample of target-domain imagery.

14. Repository

Full code, notebooks, checkpoints, and the dashboard are available in the project GitHub repository (see README.md for structure and setup instructions). A 3-minute screen-recorded demo of the live dashboard accompanies this report.